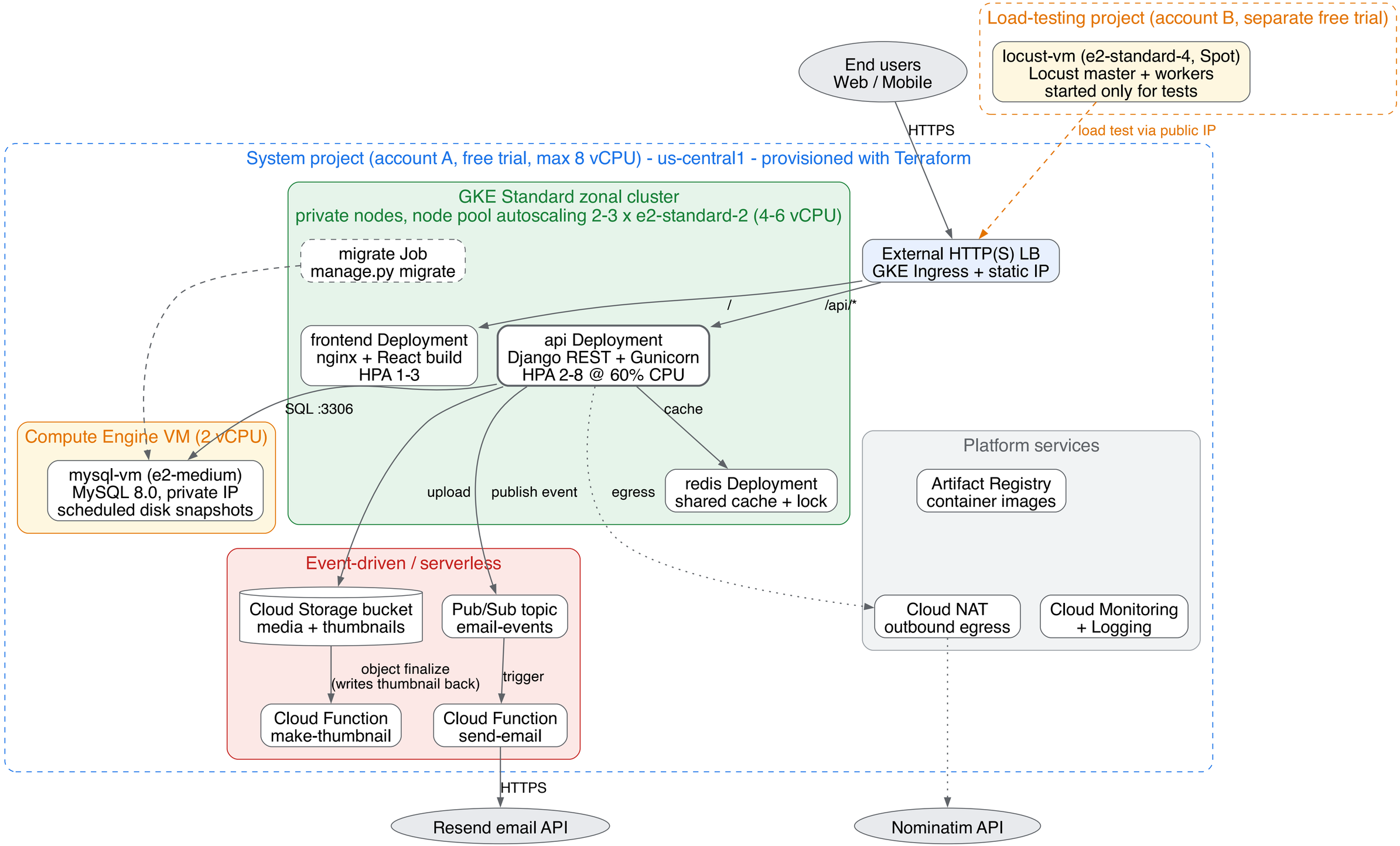
Load-testing project (account B, separate free trial)
locust-vm (e2-standard-4, Spot)
Locust master + workers
started only for tests
End users
Web / Mobile
HTTPS
load test via public IP
System project (account A, free trial, max 8 vCPU) - us-central1 - provisioned with Terraform
GKE Standard zonal cluster
private nodes, node pool autoscaling 2-3 x e2-standard-2 (4-6 vCPU)
migrate Job
manage.py migrate
External HTTP(S) LB
GKE Ingress + static IP
/
/api/*
frontend Deployment
nginx + React build
HPA 1-3
api Deployment
Django REST + Gunicorn
HPA 2-8 @ 60% CPU
SQL :3306
cache
Compute Engine VM (2 vCPU)
mysql-vm (e2-medium)
MySQL 8.0, private IP
scheduled disk snapshots
upload
publish event
egress
redis Deployment
shared cache + lock
Platform services
Artifact Registry
container images
Cloud NAT
outbound egress
Cloud Monitoring
+ Logging
Event-driven / serverless
Cloud Storage bucket
media + thumbnails
Pub/Sub topic
email-events
object finalize
(writes thumbnail back)
trigger
Cloud Function
make-thumbnail
Cloud Function
send-email
HTTPS
Resend email API
Nominatim API